

Cash Flow Forecast

Types of Cash Flows

- Organizational cashflow
 - The cash flow forecast of a company
 - Review and analyze the predicted incoming and outgoing cash for a time (year) and is often used for business and resource planning and for analyzing the financial health of companies
- Project cashflow
 - The cash flow forecast of a particular construction contract or project
 - Deals specifically with the payments due under a particular construction contract.
 - The construction contract cash flow will often inform a company's overall cash flow as they are intrinsically linked.

Project payment	Description	Accuracy
Stage payments	Pre-agreed amounts are paid at pre-set times regardless of progress on project	Provides high predictability of cash flow but low accuracy of value of works done to date
Milestone payments	Pre-agreed amounts are paid upon completion of pre-agreed targets	Provides high predictability of payments but low predictability of when they may become due.
Payment against an activity schedule	At pre-agreed timescales progress against an activity schedule is monitored and pre agreed payments are made for activities which have been completed	Provides reasonable predictability of cash flow forecast but only reasonable accuracy of value of works done to date
Valuation of works done to date on site (3rd party certification)	At pre-agreed periods, the value of work on site is assessed	Provides lowest predictability of cash flow forecast but high accuracy of value of works done to date

Stage Payments

- Stage payments are often used under design and build projects and are effectively pre-agreed values to be paid at pre-determined times.
- These are not tied to milestones and are the least accurate assessment of works carried out on project.
- There is a benefit for the project in knowing the exact amount due at any time, but this increases the risk of overpaying (or underpaying) towards the project cost.
- A valuation of works done on site could therefore be very different to a third-party certification if they are not tied to milestones

Milestone Payment

- Milestone payments are a type of stage payment but instead of being triggered by reaching a given date, payment is triggered by completing a pre-agreed milestone.
- In its simplest form this could constitute a single payment at the completion of the building, but more likely they will be tied to elements (completion of foundations, completion of frame).
- It is also possible to be very prescriptive and have milestones payments associated with sub-sections of milestone (completion of laying rebar to first floor).
- The more prescriptive the milestone payments the more accurate the assessment of works.
- However, this inevitably leads to a less accurate predictability of the cash flow.

Activity Schedule

- Activity schedule is a form of milestone payment.
- The fundamental difference is that milestone payments are made when the milestone is complete, whereas an activity schedule is assessed at regular intervals and payment made against completed activities only. (Civil construction, equipment installation, commissioning tests for a plant)
- There may be a time lag between completion of an activity under an activity schedule and recognition of that completed activity, whereas with a milestone payment a contractor can ask for certification once the milestone is complete.

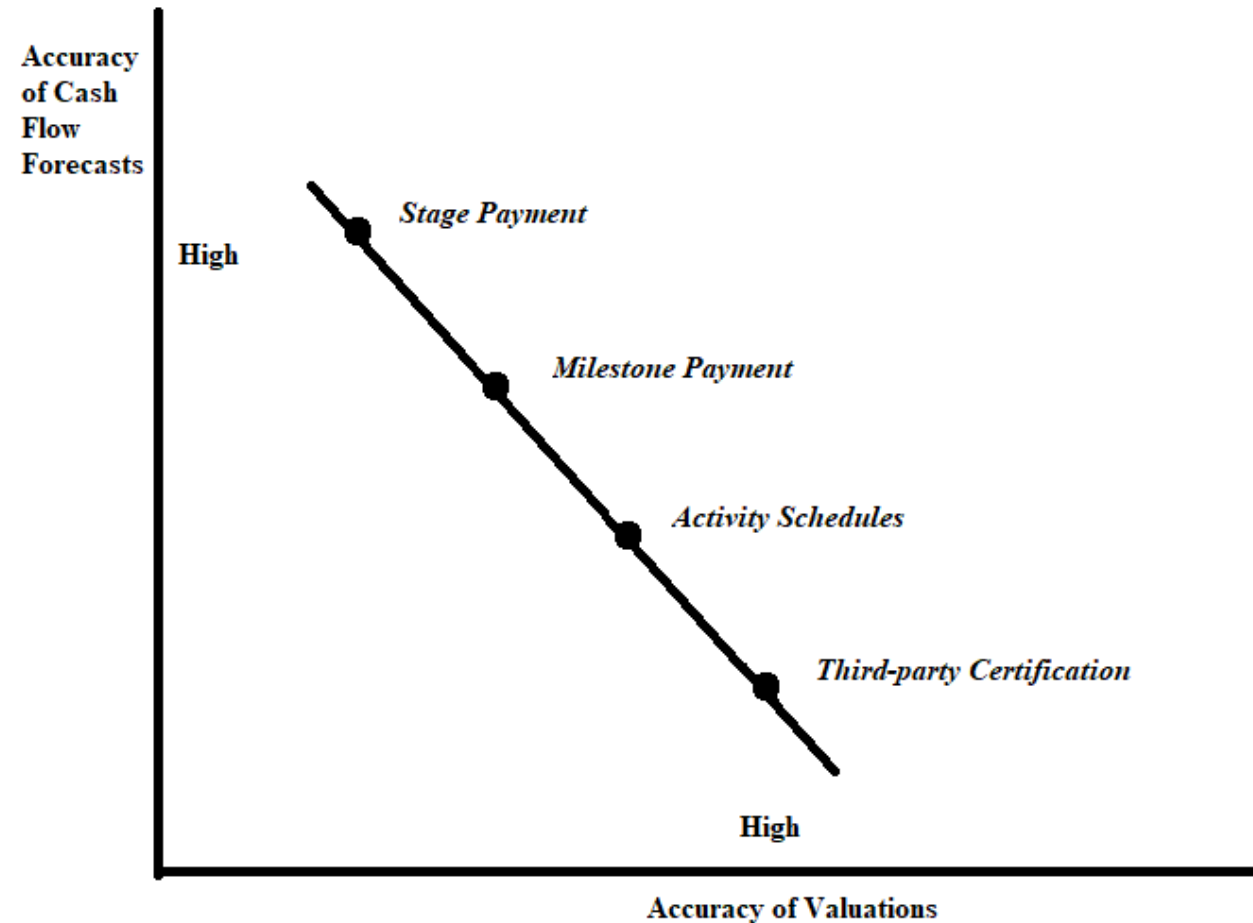
Third-party Certification

- Third party (usually the quantity surveyor) measures and values the works carried out (to date) by visiting sites and assessing the value of preliminaries, works carried out, materials on-site, materials off-site (if applicable), and any instructed variations.
- This provides the most accurate assessment of works carried out on site to date.

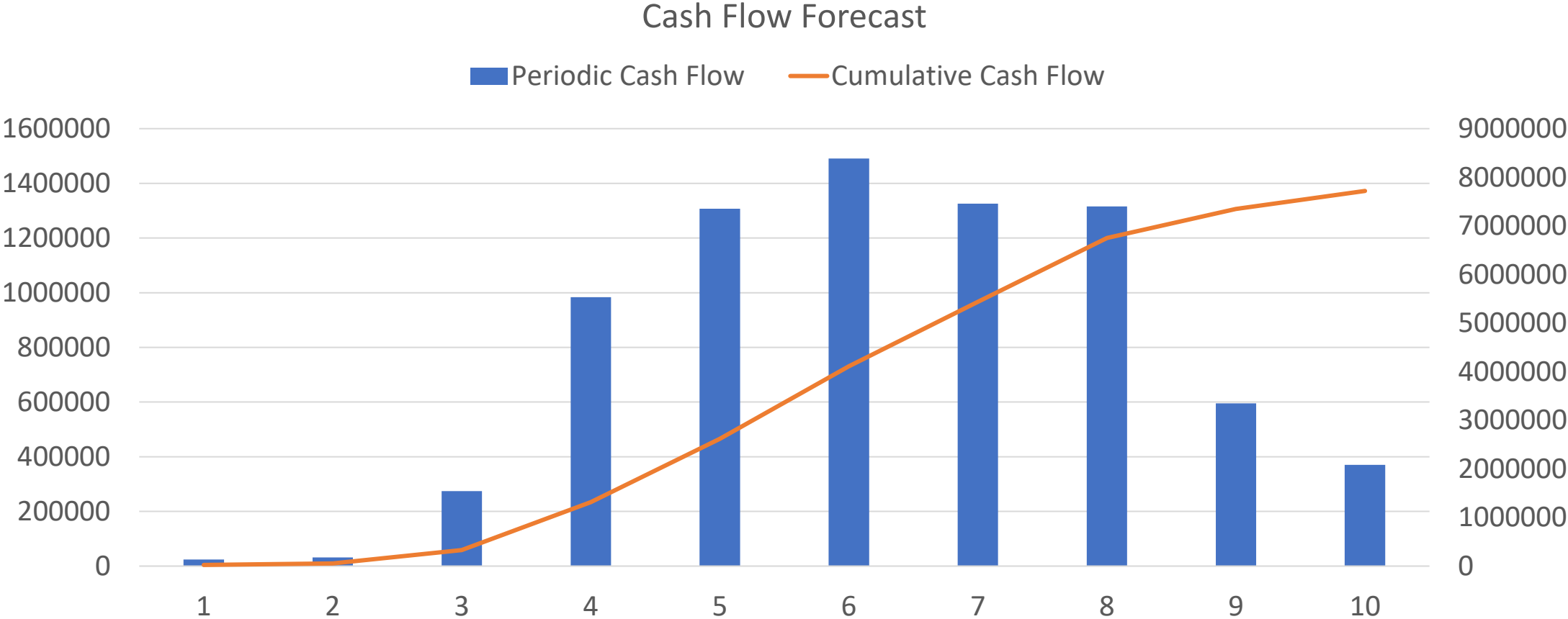
Actual Cost / Target Cost

- If an actual cost / target cost is used, and if both parties agree, then the valuations are assessed by ascertaining the actual costs plus any agreed fee percentages.
- The discrepancies between actuals and targets are offset by reviewing actual invoices.
- Most often, in several projects the project payments are not lump sum contractual costs and there is a risk that actual costs could exceed the target cost.

Trade-off between cashflow forecast and valuation



Types of Cash Flows



Need for Cash Flow Estimates

	Situation	Impact
Hard Costs	Overdrafts, unnecessary credit line drawdowns Cash in the wrong place/currency /time	Interest expenses, bank fees Vendor late fees, bank charges, FX transaction costs
Opportunity Costs	Idle cash balances Cash diverted from other parts of business	Loss of interest income; missed purchase, investment or acquisition opportunities Inefficient supply chain cycle
Risks	Lack of visibility, unforeseen currency exposures Lack of reliability in understanding upcoming cash movements	P/L impact of FX loss Excessive time/resource diversion; reduction in credit lines; potential loss of suppliers

Objectives of Cash Flow Forecast

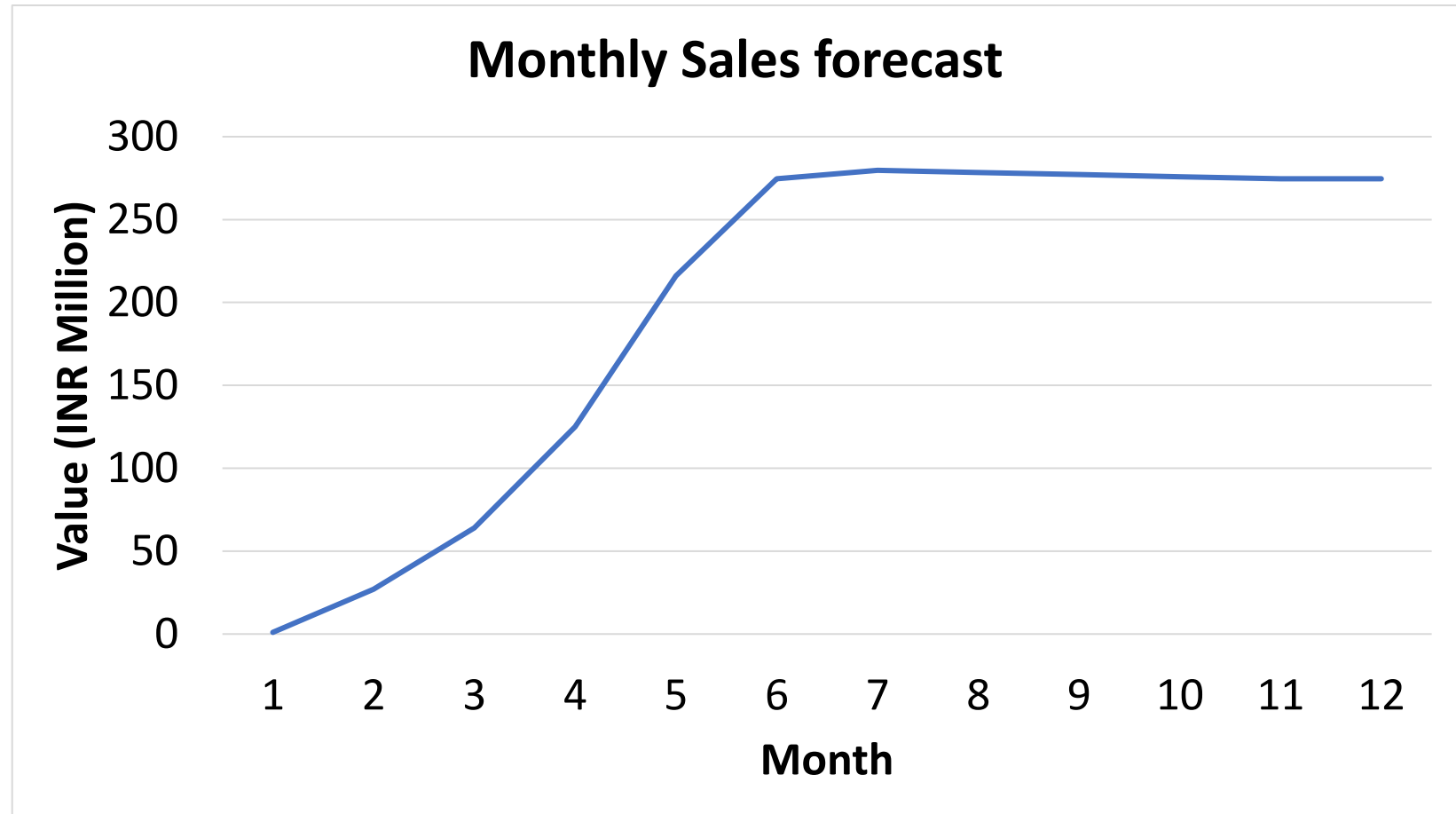
- Minimize idle cash, short term debt and overdrafts
- Optimize short-term borrowing and lending decisions
- Provide liquidity for all currency needs
- Monitor FX exposures and risks
- Manage key transactions
- Support equity debt requirements
- Provide a current view of business and financial health
- Monitor the cash generation of the firm
- Support cash deployment decisions such as cash buffers, capital spending and stock buy-backs
- Validate other forecasts such as P&L and B/S forecasts by finance and operations

Identifying Cashflows: Points to remember

- All revenue and costs are measured through cashflows
- All cashflows are incremental
- Sunk costs are not included
- Any effect / impact on other project operations are considered in cash flow analysis
- Interest paid on debt is not considered

Cash Inflows from a Project (Cumulative)

- Sales forecast for a firm (12 months)



Types of Cash Flows

- Initial cash outflows:
 - payments that occur at the inception of the project
- Operating cash flows:
 - revenues, costs, and expenses generated by the project
- Additional working capital:
 - inventories, accounts receivable, cash needed for growth that are recovered at the end of the project
- Salvage or resale values:
 - expected sales value of project machinery at end of project
- Non-cash investment:
 - use of an existing machine that is not used
- Sunk Costs
 - Cash that has already been spent before the start of the project and cannot be recovered

Operating Cash Flows

Period Ending:	Year 4	Year 3	Year 2	Year 1
Net Income	59,531,000	48,351,000	45,687,000	53,394,000
Cash Flows-Operating Activities				
Depreciation	10,903,000	10,157,000	10,505,000	11,257,000
Net Income Adjustments	(27,694,000)	10,640,000	9,634,000	5,353,000
Changes in Operating Activities				
Accounts Receivable	(13,332,000)	(6,347,000)	476,000	(3,318,000)
Changes in Inventories	828,000	(2,723,000)	217,000	(238,000)
Other Operating Activities	(423,000)	(5,318,000)	1,055,000	(283,000)
Interest expenses	47,621,000	9,465,000	(1,343,000)	15,101,000
Net Cash Flow-Operating	77,434,000	64,225,000	66,231,000	81,266,000

Example

- A firm has two bottling plants for packing in 500 ml and 250 ml bottles. Installation of a new machinery will make 250 ml production redundant. Sale proceeds from 250 ml production line is 40 million (book value 30 million).
- Capital cost of new machinery 1300 million. Other installation cost 25 million. At the end of five years utilization, the new machinery could be sold at 70 million against its book value of 20 million.
- Additional working capital requirement every year 150 million.
- Revenue expected in the first year of operation is 1500 million and is expected to record 25% growth each year. Raw material costs are 600 million and expected to grow at 20% rate each year. Other overhead costs are 500 million in the first two years and 1000 million for the remaining four years.
- Firm's income tax: 30%; Capital gains tax 40%.

Cashflows (millions)

Year	0	1	2	3	4	5
New Machine	-1325					
Sale of unused line	40					
Tax on sale	-4					
Revenue		1500	1875	2344	2930	3662
Material cost		-600	-720	-864	-1037	-1244
Overhead		-500	-500	-1000	-1000	-1000
Depreciation		-265	-265	-265	-265	-265
Profit before tax		135	390	215	628	1153
Tax on profit		41	117	64	188	346
Profit after tax		95	273	150	440	807
Machine sale						70
Tax on sale						-20
Cashflows	-1289	360	538	415	705	1122

Business Scenario

- Illustration of methods applied by firms to improve free cash flows are: volume growth or revenue growth; operating margins or profit growth; reduction in working capital cycles or operating cash flow growth; Increase in asset turnover or free cash flow growth.
- In India, all jewellery firms outsource manufacturing of gold jewellery to goldsmiths who are highly unorganised and work with outdated technologies of manufacturing, along with poor work conditions. The impact for the firm is inefficient supply chain in the form of higher wastage, high manufacturing lead times, leading to high investment in raw material/work-in-process inventory of a high value commodity like gold.
- A large jewellery firm took the initiative to provide healthy working conditions to goldsmiths who are supplying them the jewellery and train them with efficient practices (theory of constraints, lean manufacturing). As a result, over 6 years, manufacturing lead time of vendors reduced from about 35 days to about six days, reducing WIP inventory of vendors by one-sixth and improving the firm's sales with faster turnaround times and better stock availability to its customers. By introducing robotics to diamond jewellery products, the firm improved its business capacity by 10 times over 6 years and freed up hundreds of man-days of its workers.
- Make your assumptions based on the information to determine the cashflow impact for the firm.